All Data Center Management Notes - First Test

Make MONEY bro

software development, data nalytics, AI, machine learning, NOT CS professors LMFAO!

in college we learn programming languages/operating systems, algorithms, theory of cmoputations, cloud computing and security.

HOW TO MAKE MONEY!
keep up w current tech
Read research papers, everyday tech news, try latest products
(AI is latest product)
Understand peopls needs, find gaps where software can be beneficial
understand business models of tech companies

Business Model: Free (fire)
Open source:
download code, run code, don't make money but build reputation. ex linux, android, docker, firefox
Free Download: on premises, cloud
attract many users, potentially ot paid services
source code not available to public
ex: Adobe Acrobat Reader, Google Chrome (not chromium), Walmart, BestBuy
Free with purchase: on-premises
come along with other purchases (hardware) by default, but users are more interested in the software rather than hardware.
MacOS, Comma Ai, Amazon Alexa, Google Nest Hub
Free with advertisement: Cloud
free to download and use, come with advertisement in the software
gain profit from advertisements (cannot remove ads)
require subscription fee to remove
facebook, google search/maps, youtube
Free with Software tax: cloud
online stores that split money between content creators an dplatform host
need to have larger user and creator population to turn profit
apple store, google store

Business Model: Licensing
License purchase: On-premises
purchase license keys to activate functionalities of the software

one time purchase with free update/upgrade
ex: Microsoft Windows, Computer games, mobile apps
License n Upgrade purchase: cloud
purchase license keys to activate functionalities of the software
free update, paid upgrade
Often find outdated software version vulnerable (lack of security support)
VMWare workstation
Freemium: cloud
Users pay monthly subscriptoins to gain full access
free version has limitations
Evernote/overleaf
Premium: cloud
users required to pay monthly to continue using software
come with trail period
adobe creative cloud, microsoft office 365

Business MOdel: XaaS

Everything-as-a-service: cloud
Users pay for amount of services they requested, may have free tier w small amount of services
apple icloud, dropbox, chatgpt, duo security, google drive.

XaaS: Cloud Computing
Software as a service (SaaS)
Applications previded by cloud provider
google drive, microsoft 365
Platform as a Service (PaaS)
Hosted application environment
googles app engine, cloud gaming
Infrastructure as a Service (IaaS)
Virtual Machines with processor, storage, network, etc
Amazon AWS EC2

Cloud Service models
IaaS PaaS SaaS
Host Build. Consume

Benefit of XaaS
Efficiency:
make products development convenient/reliable
make development process more efficient
Cost savings:

Distributed costs and subscription-based cash flows, more cost-effective
get rid of physical IT infrastructure, lower costs related to maintence,space cooling power
Scalability:
quickly change the package and add new components experiment new features in a safe environment.
Access to the best services: Gain access to top-quality services
increased return on investment

Risks

Downtime:
internet reliability issue. Impact usability and users cannot access some resources
Performance:
Bandwidth, latency, data storagethe system can be slowed down
Limited control:
Do not accommodate customization; standard services for everyone
Locked-in:
Service dependence, difficult to migrate
Security and Privacy:
isolation failure
data security/privacy
compromised user interference and credential stolen
single point of failure

Cloud Deployment Models:

Accessible to gereral public: Amazon AWS, Google cloud
Characteristics:
shared resources: computing resources are pooled and shared among users
Scalability: offers virtually unlimited scalability ot meet fluctuation workloads
Pay-as-You-go: costs are based on usage, reducing capital expenditure
Accessibility: Services are accessible over the internet from anywhere

Private Cloud:
Characteristics:
Exclusive Access: Dedicated resources for one org
Customization: high level control over hardware + software configurations
Enhanced Security: Provides greater data protection and privacy

Use Cases:
Financial institutionsL handling sensitive financial data
Healthcare organizations: managing confidential patient records
Large Enterprises: with Stringent compliance and security needs

Hybrid Cloud:

Characteristics:
Integrated infrastructure: combines premises and cloud
flexibility: Optimizes workload placement based on performance, cost, security
scalability: leverages public cloud resources to handle peak loads.

Use cases:
Seasonal Business: Scaling resources during peak periods
Data Processing: Keeping data secure on private cloud while processing in public cloud (turbo tax)
Legacy Systems Integration: extending existing infrastructure capabilities

Community Cloud:
Cloud Infrastructure shared among several organizations with common concerns, such as security, compliance, or jurisdiction

Characteristics

Choosing a deployment model

security + compliance
cost implications
scalability needs
control and costumization
performance requirements

Cloud Migration
Cloud migration is the process of moving data, applications, or other business elements from an organizations on premises infrastructure to a cloud environment, it can also involve moving from one cloud environment to another. Known as cloud-to-cloud migration

Cloud migration strategies
rehosting
move application to the cloud as is
replatforming
modifying applications to better support cloud environment

rebuild
rewrite application from scratch
replace
retire application and replace it w new cloud-native application

Jan 21 - Day 2

Cloud Deployment Models pt 2

Soverign Cloud - Some/all of organizations data must stay within a certain national or regional geographic boundary, whether a state or country or a broader region.
many countries have adopted this long beofre GDPR
Google/Facebook does not operate in China cuz they cannot store data only within China, they odnt' want the gov to have access to their data
US also bans data to be stored in another country
A sovereign cloud is a cloud environment that helps an organization meet its digital soverignty requirements. They claim to offer, access restrictions, organizational control, compliance w gov reg or industry requirements, operational support, dedicated netowrk capacity, sophisticated encryption.

Cloud Migration

- Process of moving data, applications or other business elements from an org on premise infrastructure to a cloud environment. Also involve moving from one cloud env to known as cloud-to0cloud migration.
  -Types of cloud Migration

on premise - cloud = shifting workloads from local data centers (we talked abt this last time)
Cloud to cloud migration: moving assets between differnt cloud providers
reverse cloud migration: bringing applications and data back to on premise from the cloud.

Youtube - uses IaaS. Use IaaS - from a developer perspective. They use googles Data Centers. From costumer POV is Software as a Service.

Data Center Architecture

Data center from the inside - black LOL
warehouse w open space, every room doesnt need lighting. 3-5% of the cost is lighting. Turn on light - fire emoji
Take power put a transformer, then switchgear (connects to generator in case lose generator

from grid). After switch gear -> UPS uninterrupted power supply, take battery to give data center juice, so if grid down, battery can power, then generator can kick in. Then PDU (power distribution unit) then rack, then rack level pdu, then power every server in a server cabinet.

Power infrastructure

Transformer: Step down/up voltages
PDU: Distribute power to multiple devices, multiple tiers
power breakers: prevent overcurrent
Uninterruptible power supply (UPS): Power backup in case of outage
backup generators
automatic transfer switch (ATS): Switch between UPS and main power
Power grid -> transformer -> Switchgear (ATS) -> UPS -> PDU ->RPP -> RACK PDU.
UPS -> also UPS Batteries.
Switchgear (ATS) -> also generator

Thermal system

Everything in datacenter produces heat
Globablly, data centers consume 205 terawatt-hours (TWh) annually
cool aisle/hot aisle
more discussions later

Emergency System

weather forecast/monitoring
Equipment stress: Extreme temperatures or weather changes (heat waves, cold snaps) Place additional stress on mechanical systems like chillers, generators, UPS batteries, high humidity aslo impact performance
Disaster Prepardness: Storms, hurricanes, floods, or blizzards can disrupt power lines, create transportation issues for staff and potentially threaten physical security
cable damage in severe conditions: strong winds, heavy snow, flooding can damage fiber cables or cause local network outages
supply chain and logistics: site access is limited and staff presense is low

Emergency System

Fire and life safety system
Protecting people, exit infrastructure as quickly as possible in case of fire, fall detection, EMS notif
Protecting Critical Infrastructure: robust fire suppression system
regulatory and insurance requirements
cost and efficiency and reputation
equipment
VESDA: Very early smoke detection apparatus
HSSD: High sensitivity smoke detection

Security system

Layers of physical Security
Perimiter: Fences, cameras, detterrence
Facility exterior: badge control, CCTV Loading zone control
Facility Interior: Visitor checkin, airlock, biometric scanner
Server room: Authorized personnel only, cabinet rack lock
Other rooms (power room, AC room): Restricted access zero trust

Control/Monitoring System

IDCMS: integrated Data Center Management Suite
software suite or a combination of software and hardware tools, designed to centralize and streamline the monitoring and management of critical data center components
IT equipment: Servers, storage arrays, networking gear
Facilities Infrastructure: Power distribution units, cooling systems, generators, UPS
Environmental controls: Temp, humid, air flow, fire detection/suppresion

Control/Monitoring system

Purpose:
Reliable + Efficient operation
Overall cost reduction and risk mitigation
capabilities to expand IDCM approach and functionality over time

Available, safe and secure infrastructure
fast/better decision makign due to transparant, personalized dashboards

On-going research (whats next)

Climate change + impact on DC:
moving beyond disaster recovery
Climate change adaption extends this to long-term architectural and operational shifts
AIOPS:
USing AI and big data analytics to automate and enhance IT operations
Automate Incident diagnosis and resolution, reduce downtime, improve resource utilization
LLMS generative techniques for log anomaly detection and operational diagnostics
Digital Twins:
real time virtual rep of physical systems
"What if" simulations, scenario analysis, predictive forecasting, performance optimization + resilience planning

Jan 26 - Day 3

Power Supply Unit -
Converts high AC (alternating Current) from a wall outlet into the stable, low voltage DC(direct current) needed to power all internal parts

- AC to DC conversion
- Power distribution" Delivers these convereted power to every part of PC
- Protection: Includes built in safeguards against power surges

Power Budget

- in early 21 century power budget of a rack or blad server psu was in the 200W to 300W range
    - CPU power consumption: 30-50W
- Todday Server PSU power budget is > 3,000 W
    - CPU power consumption: 200W - 300W
    - GPU power consumption: 500W-800W per chip (For ML and AI computation)
- PSU power budgets are limited by their AC inlet current rating
    - 230V x 20A(16A) = 4,600W (3,680W)

- NEC (National Electrical Code) requires that the sustained current shouldn't exceed 80% of a breakers rating

PSU Ratings

- from physics calss we know that Power(W) = Volatage(V) x Current(A)
  - True in Direct Current, wrong in alternate current
  - va = wats/power factor
  - in power management, we use VA not W
- 3,680W at 230V with 80 plus titanium PSU
  - 3,680W/94% = 3,914VA
  - 3,914VA / 230V = 17Amps

Power Distribution Unit:
(fat phoebe rn) but there are ofc other versions

Rack based

- Used to effeciantly distribute power in a rack w multiple outlets and range of intelligent features to control the power distributed to IT devices
- Rack Automatic transfer switches (ATS) provide automatic transfers froma primary to secondary source to power critical equipment without interuption
  CabinetBased
- Power distribution racks (PDR) are typically seen in larger high density data center environments to provides space saving power distribution in flexible design
- PD cabinets of large PDUs take incoming power and distribute it to an indivudual rack or groups of racks
  Other:
- Busways provide flexible overhead power distribution where change and adaption are important

Typical PDU configuration w UPS models
PDU = big power strip, Traditional installation,

Maintenance bypass = Wall -> PDU. Some outages, bypass will switch to UPS
Dual feed Dual UPS
Dual feed single UPS

1-Phase vs 3-phases
1phase power deliver

- Single sine wave voltage on a power cable
  3-phase power delivery
- same voltage and current can support (square root of 3) times the power
- Thinner cables and bigger plugs, requires less PDU compared to a 1-phase
  PDU limits the total power capacity of a rack
- We can fit as many machines as possible under that PDU
- Otherwise, circuit breaker will be triggered :(

DC Power Distribution
Reduced Conversion Losses:

Improved Reliability

High Power density:
DC systems especially at higher voltages can offer high power density w

Cost:
CD power distribution require more expensive equipment
Less devices + easy installation = at much as 30% less cost

Key elements lead to downtime

- UPS battery failure
- UPS capacity exceeded
- Accidental EPO/Human Error = higher n higher
- UPS equpment failure = 49 %
- water Incursion = 35%
- Heat Related/Crac failure = 33 %
- PDU/Circuit Breaker failure = 33%

Redundancy
N = No redundancy, only what you need
N+1 = One additional (spare) Module over the required number of modules
2n = Twice the number of modules needed (fully duplicated system)
2N+1 = A fully duplicated system with one extra module for even higher reliability

Datacenter uptime tier classification

- Datacenter downtime cost
  - Avg cost of downtime is 5,600 dollars per min
  - 40% of enterprises said the cost of an hour of downtime can range from $1 million to more than 5 mil and that does not include legal fees fines or penalties

- Uptie tier classifications
- DC Tier 1 Uptime = 99.671% or less than 28.8 hours of downtime per year
- DCT2 Up = 99.741 or less than 22 horus of downtime per year
- DCT3 UP = 99.982 or less than 1.6 hours of downtime per year
- DCT4 UP = 99.995% or less than 26.3 min of downtime per year.

DCM Jan 28 - day 4

DataCenter Power Utilization

- it equipment (30.5%)
- Electricity (24%)
- INfrastructure (37.5%)
- Other POEX (8%)

Cooling System Design

- "Grab a jacket, we are going in the datacenter"
- ASHRAE 9.9 recommends inlet temperature to be 18-20 degree C and 20%-80% relative humidity (common practice 18-27 Degree C)

CRAC (Coputer Room Air Conditioner)

- CRAC units use a direct expansion refrigeration cycle involving a compressor and refrigerant to cool conditioned air inside a data center
- Key characteristics:
  - self contained system
  - Uses Electromechanical cooling cycle
  - often requires external condensing (air vs wet)

Thermal Managemetn (popular way to do cooling.)
cool aisle hot aisle

CRAH (Computer Room Air Handler)

Other types of cooling system

- Indirect Water Cooling
  - Reer Door Heat Exchangers (RDHx): function like radiators affixed to the back of server racks, equipped wit hcoils to faciliatte heat exchange with circulating water

    - no hot aisle recirculation or cold air bypass, no need for hot aisle containment or hot air return
- direct water cooling
    - circulate coolant directly on top of processors
    - creates a complex heat exchanging loop composed of pump piping and manifolds fluid couplings and terminators.
      Immersion Cooling
      Submerge hardware into a bath of flowing liquid, all components are in direct contact with head transfer liquid
      Liquids should be non-flammable and non-conductive
      Efficient, but take too much spaces (servers cannot stack)

Overheating
4 temperature measurements
server inlet/outlet temperature
AC inlet/outlet temperature
What can cause server Inlet temperature to
Tiles get cracked somehow, crac doenst output the right temperature, hot air, flows back to inlet, or someone is servicing the servers and the person standing in front of the server, blocks cool air (their body is not cool).
What would you do if you feel cold in a datacenter
stand in hot isle, open the door in hotisle,

DataCenter Power Consumption
Electricity and power usage
Computing Hardware is typically the largest consumer of electricity, high-performance hardware can be particularly power-intensive

How t ocalculate - Power usage Effectiveness (PUE)
The datacenter pulls 1,000 kW from the grid with the following distribution
Total facility power = 1,000 kW
Total IT load power = 2,000 * 37% = 370kW
PUE = 1,000 /370 = 2.7 (pretty bad!)
How to decrease PUE
Increase power delivery efficiency
reduce cooling, power conversion, and lighting

Example: Failed Projet
GOogle Barges - floating dtaa centers in the water; using wave energy for power and sea water for server cooling.

Example: future project
Lonestar Lunar: moving datacenter to the moon.

Greenhouse Gas Emissions
Scope 1: Direct emissions:
backup generators, refrigerants, vehicle fleets
0.2-0.5%
Scope 2: indirect emissions (purchased energy)
Purchased electrcity, district cooling/heating
31-61%
Scope 3: indirect emissions (value chain)
supply chain, building materials, shipping

Electronic waste (E-Waste)
Global ewaste will reach 74.7 metric tons by 2030 20% is properly recylced
Challenge:
Lifecycle management of servers and equipment
hardware disposal and repurpose
Junkyard data center: repurposing discarded ....

Feb 2nd Notes - Day 5

Electricity + Power Useage
computing hardware (servers, storage, networking) is typically the largest consumer of electricity. High-performance hardware. can be particually power intensive
cooling systems - often account for second-largest share of power consumption in a data center
Power usage effectiveness (PUE)
PUE is a commonly used metric to gauge how effficiently a data center uses energy
a perfectly efficient data center would havea PUE of 1.0, meaning all energy goes directly to IT

WUE - water usage effectiveness
rely on water for cooling. 100,000 liters of water per day 50,000 kilowatt hours of energy your WUE calculation would be 100,000/50,000 = 2 Litersrs/kWH
average WUE across data centers is 1.8 L/kWH; amazon reports that its WUE is 0.15 L/kWH
amazon uses on site sewage treatment plants at 27 logistics sites to recycle greywater for toilet flushing nad irrigation, which is expected to save 279 million liters of water per year.

Test - 02/09 EVERYTHING INCLUDING AND UP TO THIS LECTURE T/f multiple choice, and short response

Virtualization:
Cloud computing - fire

Virtualization is a process of creating virtual versions of physical components- computing hardware, storage devices, networks, or even entire OS's. It allows one physical system to run multiple simulatied environments or systems
key tech in cloud computing
Turn physical hardware into agile, on-demmand resources that can be quickly scaled, moved, or replaced.
key concepts
Abstraction: separating the underlying hardware from software running on it.
resource pooling: multiple virtual environemnts share the same physical hardware resources
real world example: USPS

WHyyyy virtualization:
Efficient resource utilization:

- Consolidation: By running multiple VMs on a single physical server, data centers reduce the number
- Scalability:

Simplified management and maintenance:

- centralized administration:
- Rapid Provisioning: because everythign is softwre, it runs fast
- Automated workflows: Everything is software, user can actually go in and do it by themselves. They can set up automatic flows (software) user/data center manager. If i had two instances they need to talk, so if user has 2 instance, i'll configure

Improved fault tolerance and disaster recovery:
Live migration:
High availibility:
Simplified backouts and restores:
Security enhancements:
Isolations:
Segmentation:

What can be virtualized

- everyting
- Compute
    - create virtual CPUs and memory pools so multiple vms can share the same physical hardware.

    - improves resource utililzation and flexibility
- network
    - software-defined networking (SDN), cirtual switches, routers, firewalls
    - Enables dynamic network segmentation, isolation and easier provisioning
- Storange:
    - logical volumes (LUNs) abstract physical disks or SAN devices
    - lets you pool multiple physical disks into a single Virtual storage resource

What else can be virtualized

Desktops

- virtual desktop infrastructure (CDI) delivers entire desktop OS environments over a network
- Centralizzes desktop management and enhances security
  Appplications
- Application virtualiztion runs apps in isolated "bubbles" separate from the underlying OS
- Containerization (e.g Docker) Virtualizes at the OS level for lightweight deployments
  GPUs and Specialized Hardware
- GPU virtualization shares high-performance graphics or compute

Desktop Virtualization

GPU virtualization

What can be virtualized?
Data/Databases

- Data virtualization abstracts data from disparate systems, presenting a unified view. (hello world.txt to main director (from users perspective thats where that data is stored) but behidn the scene the DC can take the data and store it somewhere else)
- Database virtualization uses snapshots or clones to spin up test or development environments quickly
- and abstract layer that allows you to access data from multiple different sources (databases, files, cloud) as if it were a single place, without moving the data.
  Operating systems:
- OS level virtualization (containers, zones, jails) provide multiple user-space instances on one kernel.
- Each instance is isolated but shares the underlying operating system
  Entire Data Centers:
- software defined data centers (SDDC) virtualizes compute, storage, and networking at scale.

- enables automated provisioning and high-level orchestration across all resources

Datacenter Virtualization:

Putting everything together: cloudlab
CloudLab is a testbed for research and educaiton in cloud computing:

- provides more control, visibility, and performance isolation than a typical cloud environment
- support work on cloud architectures, distributed systems, and applications
  Cloud environment
- simplifies selection of hardware configuration, creation of custom images, automation for software installation and configuration
- Cloudlab staff take care of construction, maintenence, operation, etc. of the facility. Letting users focus on their research
  Not a traditional cloud environment
- allow users to build entire clouds from the "bare metal" up (hardware-level user isolation)
- Give users unmediated "raw" access to hardware to run fully observable and repearable experiments.
- allow users to see, instrument, monitor, and modify all levels of investigated

Cloud Lab

- use profiles
  - the topology (nodes and links)
  - Hardawre classes and storage images
  - software to run on nodes
  - network configurations
- On demand profiles: temp experiments (hours to days)
- Persistent profiles: long lived research environments (weeks-> months)

Introduction to research and research papers:
Technology keeps evolving

- apollo guidence computer (moon lander) had 0.043 MHz CPU and 64K memory
- iphone introduced 18 years ago

Feb 4th Notes - Day 6 notes

Computer Networks in 75 min

What is the internet?

- Vast, interconnected system of computer networks that span the globe
- Communication infrastructure
  - enables distriubted applications
  - WEB, VoIP, email, games, e-commerce, etc.
- Internet "Network of networks"
  - loosely hierarchial
  - runs by protocol (commmon language that everyone understands)
  - two people able to understand each other

History

- 1960-1970s ARPANET, funded by US depeartment of defense sites.
- 1980s-1990s: more networks joined, world wide web invented
  - int is underlying network of computers/connections
  - WW

How int Works:

- Data packets: information traveling across internet is broken into small chunks called "packets" Each contains
  - A portion of the actual information being sent (part of email or piece of video)
  - Address information (like a return address on an envelope) telling routers where to send the packet
- server + clients:
  - A server is a computer that provides services or resources to other computers - like hosting websites, files, or email accoutns.
  - a client is a computer/device that accesses those services
- Each packet includes
  - source destination IP, protocol
  - time to live (TTL)
  - data

Internet protocol stack

- application layer: FTP, SMTP, HTTP
  - supporting network applications
- Transport layer: TCP, UDP
  - process to process data transfer

- Newtork layer: IP
  - routing of datagrams from source to destination
- Link Layer: Ethernet, wireless
  - data transfers between neighboring netowrk devices
- Physical layer
  - bits on wires
- Encapsulation: Each layer wraps data with its own header

Physical Layer

- Wires, fibers, cables, radio waves: two devices to communicate with each other

How to enable multiple devices to communicate using a single wire:

Link Layer:

- Link Layer (MAC layer)
  - Logical netowkr component used to interconnect nodes in the network
  - allow devices on a single wire to communicate with each other (LAN)
  - Bridging the network nodes (IP addresses) with phycial devices (routers/switches)
- Medium access control (MAC): coordinating access ot the shared medium
  - Unique hardware identifiers for devices, although modern software can report random MAC addresses
  - format: 48 bit address
- Protocols:
  - IEEE 802.3

Network Layers

- IP addressses to idnetify devices in other local area networks
- key functions of network layer
  - forwarding: move packets from router's input to appropriate output
  - Routing: Determine the route taken by packet from source to destination
- Data plane
  - local, per-router function
  - determines how arriving area is forwarded
- Control Plane
  - Network-wide logic
  - Determines how data is routed from source to destination
  - two approaches: Routing algorithms, software-defined networking (SDN)

Transport layer

- user datagram protocol (UDP)
    - point to point one sender, one reciever
    - barebones protocol: Lost or out of order
    - connectionless: no handshaking
    - fast
    - can be made reliable (user efforts)
- Transmission control protocol (TCP)
    - Point-to-point: one sender one reciever
    - reliable in order byte stream
    - popelined: congestion and flow control
    - full duplex data: bi-directional data flow
    - connection oriented: handshaking

Application Layer

- In cloud, users may have control over the application layer
- IaaS and PaaS
    - users specify desired protocols depending on application
- SaaS
    - cloud providers specify desired protocols

Scale of datacenter

- datacenter is large facility that hosts ~100k servers with
    - 100s of petabtes of storage
    - 100s of terabits of bandwidth
    - 10-100mw of power
    - cost $1-4 billion to build
- Datacenter traffic growth

Switch vs router

- Switch
    - layer 2 devices (link layer) to forward data within a single network
    - operate based on MAC addresses, self learning what devices are connected to which part
- Router
    - Layer 3 devices to route packets between different networks

Top of rack architecture

- Rack of servers
  - High end commodity servers
  - top of rack switches
- Modular design
  - pre configured racks
  - power, network, and storage cabling

Why ToR?

- Less clutter
  - Connections are within each server cabinet (No messy cables)
  - save cost on cabling
- Flexibility in placement
  - can be placed anywhere in the cabinet(shorter cabling)
- Easy to rearrange
  - each cabinet is self-contained
  - quick rearrangement within datacenter
- efficient troubleshooting

Why. not ToR?

- Increased nubmer of switches
  - required one or more switches per rack, leading to higher total device count

End of row architecture

- Rack of servers
  - high end comodity servers
  - no switches in rack
- Modularize by funtionalaity
  - servers and storage in rack
  - switches in seperate rack

Keep building up

- Access layer switch
  - Commodity switches
  - commercial level (more expensive than sonsumer switch)
- Aggregation switches1.

    - datacenter level switch
- Core switches/Routers
    - datacenter-level switch